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How chatbot anthropomorphism shapes customer purchase decisions: Evidence from a field experiment on a cosmetic E-commerce retailer

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ABSTRACT

This study investigates how chatbot anthropomorphism influences customer purchase behavior through a real-world field experiment on a Japanese cosmetic e-commerce website. Using a 2 × 2 framework of form realism (human-like/cartoon appearance) and behavioral realism (warmth/competence response styles), the experiment examines how chatbot anthropomorphic designs influence one-time and subscription purchases over the short and long term. The findings show that low form realism (cartoon appearance) combined with high behavioral realism (warmth response) significantly enhances subscription purchases, aligning with the expectancy-disconfirmation theory. In contrast, chatbots with high behavioral realism (competence response) are more effective for one-time order customers, reflecting that cognitive cues are preferred in the exploration phase of customer–seller relationships. The study also reveals contrasting temporal patterns: while the novelty effect of chatbots diminishes over time for one-time purchases, subscription purchases exhibit no comparable decline, suggesting a persistent impact of chatbots in the maturity phase. These findings enhance the external validity of prior laboratory research and clarify the conditions under which low form realism can outperform human-like designs. The study demonstrates that consumers' responses to chatbots evolve across repeated transactions, highlighting the importance of considering relationship phases and timeframes. No previous study has simultaneously examined both dimensions of realism; this study addresses this gap, offering theoretical insights into the effects of chatbot anthropomorphism and practical guidance for tailoring chatbot appearance and communication style to different customer segments.

1. Introduction

Artificial intelligence (AI) technology has advanced rapidly in recent years, thereby increasing the sophistication of chatbots on various online platforms. Recent studies have shown that 62 % of customers prefer interacting with a customer service chatbot over waiting for a human agent and 69 % are satisfied with their most recent chatbot interactions (Bilan, 2024; Crolic et al., 2022; Miao et al., 2022).

For e-commerce retailers, chatbots have become critical touchpoints in customer relationship management, attracting customer interest, providing personalized recommendations, and driving sales. The chatbot usage rate among Japanese e-commerce retailers is approximately 13 % and is as high as 18 %, particularly among younger consumers (J. D. Power, 2025). However, numerous customers express skepticism or dissatisfaction toward the chatbots they interact with. Research has attributed this dissatisfaction to various factors, such as misunderstood questions, irrelevant responses, and poor integration with human service agents (Shumanov and Johnson, 2021), as well as chatbot communication behaviors that fail to foster customer relationships (Van Pinxteren et al., 2020). Accordingly, designing chatbots to enhance their reliability is a critical managerial challenge for firms seeking to improve operational efficiency through chatbot use in e-commerce.

To address these challenges, studies have focused on social cues in designing chatbots' appearance and interaction style with customers (Chaves and Gerosa, 2021; Feine et al., 2019). Designing human-like chatbots can improve customers' perceptions of both the chatbots and the company (Araujo, 2018; Go and Sundar, 2019). Furthermore, human-like chatbots can boost customers' purchase intention (Han,

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2021; Roy and Naidoo, 2021; Yen and Chiang, 2021). Miao et al. (2022) summarize these research efforts and propose a classification framework that distinguishes chatbots' anthropomorphism into a 2 × 2 taxonomy of anthropomorphic dimensions: form realism and behavioral realism. They also hypothesized that cartoonish chatbots (low form realism) with human-like interactions (high behavioral realism), such as AI chatbots, may outperform the other combinations. Following their framework, this study uses these dimensions to design experimental conditions.

Nevertheless, existing studies have yielded mixed results owing to their diverse contexts and different experimental settings for chatbot taxonomies (see the summary of related research in Table 1). First, studies on form realism offer mixed evidence. Han (2021) estimated a structural equation model to investigate how chatbot anthropomorphic design cues affect purchase intention, employing social presence (the feeling of being with another person) as the mediating variable. Their results revealed that anthropomorphic design positively affected young consumers' perceptions, making chatbot interactions more enjoyable and enhancing consumers' willingness to pay (WTP). Furthermore, Sengupta et al. (2021) examined how a human-like design influences the anticipation of embarrassment. They found that while anthropomorphism can increase user engagement, it can also help avoid potentially embarrassing scenarios. Pizzi et al. (2021) examined how chatbots' form realism and initiation mode affect consumer reactance and satisfaction: despite initial consumer reactions, nonanthropomorphic, automatically activated assistants led to higher satisfaction. Finally, Yao et al. (2023) investigated how a human-like appearance influenced customer reactions to intelligent customer service avatars following service failures. They found that competent chatbots raised expectations, leading to disappointment when failures occurred; however, warmth chatbots helped alleviate disappointment by fostering a social connection, and a cartoonish appearance mitigated negative reactions by reducing the impact of these failures.

Second, in the literature on behavioral realism, Bergner et al. (2023) investigated how AI-based conversational interfaces (turn-taking, turn initiation, and grounding) affect consumer–brand relationships. These interfaces enhanced the perceived humanness of the AI chatbot, enhancing consumer–brand connections and positive brand behaviors, such as increased loyalty and WTP. Chen et al. (2021) supported the cognitive fit theory, finding that matching a chatbot's language style with the customer's shopping task (searching or browsing) led to better decisions. They also determined that socially oriented chatbots are generally preferred. Hildebrand and Bergner (2021) examined trust formation between humans and chatbots, focusing on dialogue-based interactions with nonanthropomorphic chatbots, and found that conversational chatbots facilitated the formation of user trust and enhanced user perceptions of financial firms. Moreover, Kull et al. (2021) studied initial chatbot messages and brand engagement, revealing that chatbots using a warm, friendly tone in their initial messages fostered stronger consumer–brand connections and increased brand engagement compared with those using a competent tone. Schanke et al. (2021) conducted a field experiment with a US-based clothing retailer and showed that chatbots human-like features—such as humor, communication delays, and names—increased transaction conversion rates and made customers more receptive to offers.

To the best of our knowledge, no previous study has examined form and behavioral realism simultaneously. Two recent meta-analysis studies have addressed both dimensions of realism; however, their findings remain inconsistent. Blut et al. (2024) demonstrated that anthropomorphism in virtual agents (VAs) significantly affected agents' intentions and actual use. Specifically, form realism (avatar-based VAs) amplified the negative effect on emotional responses and usage intention, while behavioral realism (intelligent VAs) produced both positive and negative impacts. In contrast, Yanxia et al. (2024) examined how chatbot anthropomorphism influences multiple stages of customers' journey (consumers' attitudes, usage intentions, purchase behaviors, satisfaction, and continuous purchase behavior). They found that form realism (social presence cues), behavioral realism (emotional presence cues), and their combination significantly enhanced perceived

Table 1
Summary of related research.

Study	Brands/Products	Methods	Form realism	Behavioral realism	DVs	MVs (context)
This study	Cosmetic e-commerce brand	**Field experiment**	**Yes**	**Yes**	Short-term/long-term purchase behaviors	**Timeframes, order types**
Blut et al. (2024)	–	Meta-analysis	Yes	Yes	Technology use: behavioral intention, usage	Cognitive/emotional responses
Yanxia et al. (2024)	–	Meta-analysis	Yes	Yes	Attitude, intention to use, purchase, satisfaction, continued purchase/use	Service result/type, authenticity, country, students
Miao et al. (2022)	–	Review	Yes	Yes	Likelihood to purchase, avatar using intent, WOM	Online experience, media channel, brand relationship
Yao et al. (2023)	Fictitious brands	2 Lab experiments	Yes	No	Willingness to interact, disappointment	Service failure
Han (2021)	Order pizza	Lab experiment	Yes	No	Purchase intention	Perceived enjoyment, social presence
Pizzi et al. (2021)	Tariff, car rental	2 Lab experiments	Yes	No	Choice satisfaction, reaction, difficulty, confidence, performance	Assistant initiation: user-initiated vs. system-initiated
Sengupta et al. (2021)	New product, online delivery	4 Lab experiments	Yes	No	Likelihood of using a search aid	Social presence, topic embarrassment
Bergner et al. (2023)	Fictitious brands	4 Lab experiments	No	Yes	Brand outcomes: recommendation acceptance, WTP premium, brand advocacy, brand loyalty	Interface humanness, brand intimacy, need for understanding
Chen et al. (2021)	Facebook Messenger chatbots	Online field experiment	No	Yes	Cognitive fit, task complexity, decision quality, decision effort	Decisional guidance: suggestive vs. informative
Hildebrand and Bergner (2021)	Risk assessment, portfolio selection	4 Lab experiments	No	Yes	Firm perception, recommendation acceptance, portfolio allocation	Affective trust
Kull et al. (2021)	Fictitious travel brand, Bank	3 Lab experiments	No	Yes	Brand engagement	Brand-self distance, brand affiliation
Roy and Naidoo (2021)	Fictitious hotel brand	4 Lab experiments	No	Yes	Attitude, purchase intention	**Time orientation**, social perceptions of brand
Schanke et al. (2021)	E-commerce clothing	**Field experiment**	Name only	Yes	Treatment count, cash offer, conversion	–
Shumanov and Johnson (2021)	Telecom	Field data	No	Yes	Chatbot engagement (duration), purchase behavior	Consumer personality: extrovert vs. introvert

anthropomorphism. In particular, mixed cues produced the strongest positive effects on users' behavioral outcomes. In contrast, Yanxia et al. (2024) found results that contradicted the propositions of Miao et al. (2022); however, they did not explore the interaction effect between form and behavioral realism in detail. Therefore, this study examines the interaction effect between these dimensions of chatbot anthropomorphism by combining two levels of form realism (human-like vs. cartoon appearance) with two levels of behavioral realism (warmth vs. competent response). In accordance with Miao et al.'s (2022) propositions, this study's findings indicate that low form realism (cartoon appearance) combined with high behavioral realism (warmth response) significantly enhances purchase intent. Furthermore, among customers with a longer relationship with the retailer, the cartoon/warmth chatbot exhibits a temporal effect, with its advantage diminishing over time, unlike among customers with a shorter relationship.

An additional research gap exists in the literature; most studies have conducted laboratory experiments, emphasizing the need to explore consumers' interactive experiences with chatbots based on first-hand data collected in real-world field settings (Burton et al., 2020; Castelo et al., 2023; Mende et al., 2019; Van Pinxteren et al., 2020; Yanxia et al., 2024). Prior work has primarily relied on hypothetical scenarios; in contrast, this study conducts a field experiment in which we randomly assign customers to one of four types of anthropomorphic chatbots in a randomized controlled trial (RCT). Findings based on actual customer behavior enhance the external validity of existing theoretical claims and identify the boundary conditions under which low form realism chatbots outperform highly human-like designs. Moreover, by analyzing repeated transaction data from the field experiment, this study examines how the effectiveness of chatbots on customers' long-term purchase behaviors evolves over different customer–seller relationship phases. This dynamic perspective enriches existing theories by revealing that customers' responses vary over time.

In summary, the purpose of this research is to investigate how the interplay between form and behavioral realism in anthropomorphic chatbots shapes customers' purchase behaviors. By conducting a large-scale randomized field experiment on an e-commerce site, the study evaluates the effects of anthropomorphic chatbot designs on short- and long-term outcomes across different phases of the customer–seller relationship. This study offers the first real-world evidence on the interaction effects of chatbot anthropomorphism, demonstrating how these effects evolve over time, thereby identifying boundary conditions for existing theories. These insights also provide actionable guidance for practitioners designing chatbots in increasingly AI-driven retail environments.

The remainder of this paper is structured as follows. Section 2 provides an overview of the theoretical background and presents this paper's uniqueness and hypotheses, and Section 3 presents the research methodology and provides details of the empirical study. Section 4 analyzes the data and presents the results, while Section 5 discusses theoretical and managerial implications. Finally, Section 6 concludes with a summary of the findings and suggests future research directions.

2. Hypotheses development

According to Miao et al. (2022), research on consumers' responses to anthropomorphic chatbots in the literature, primarily laboratory experiments, has produced mixed results. To resolve these contradictions, Miao et al. (2022) proposed the theory of avatar marketing and expected that a positive disconfirmation would occur with chatbots that combine low form realism and high behavioral realism (as stated in their Proposition 2). According to the expectancy-disconfirmation theory, customer satisfaction increases when results exceed expectations and decreases when actual outcomes fall short of expectations (Oliver, 1980). For example, Yao et al. (2023) confirmed that a cartoon chatbot with a "warm" appearance effectively mitigated customers' negative emotions and reactions following service failures. However, this argument has not been validated in field experiments. This study expects the actual field experiment framework to yield valid results; therefore, we posit the following hypothesis:

H1. Combining low form realism (cartoon appearance) and high behavioral realism (warmth response style) in an anthropomorphic chatbot is more effective than other combinations.

Miao et al. (2022) also predicted a moderation effect by customer–seller relationship phases in Proposition 11 of their theory of avatar marketing. They anticipated that customers prefer cognitive responses from chatbots in the exploration phase, but such responses remain suppressed in the maturity phase. In contrast, the effects of chatbots' affective responses are enhanced in the maturity phase but suppressed in the exploration phase. In this study's field experiment, the customer–-seller relationship phases are determined according to two purchase options of the Japanese cosmetic e-commerce site: (1) in the exploration phase, customers buy products each time (one-time orders), and (2) in the maturity phase, they make periodic purchases via subscriptions. Subscription orders, such as Amazon's subscription and IPSY's subscription box service "BOXYCHARM," enable customers to choose from cosmetic products at first; these products are then delivered periodically at discounted prices according to their preferred time intervals (Bhatt et al., 2021; Li et al., 2023b). On this e-commerce site, one-time orders are more expensive than subscriptions and are often selected by customers who want to make a trial purchase or are undecided about committing to continued use of a product. Thus, one-time purchase customers are assumed to be in the exploration phase.

Customer–seller relationship phases could vary based on the ordering options. Research on retailers' subscription-based online services (SOS) has shown that customers with high satisfaction and attachment toward retailers are more likely to adopt SOS (Kerschbaumer et al., 2023). Furthermore, hedonic aspects of their products and services significantly influence customers' attitudes toward subscriptions and purchase behavior in the fashion and beauty SOS (Bhatt et al., 2021; Ramkumar and Woo, 2018; Sivathanu, 2018; Woo and Ramkumar, 2018). For example, Baek and Kim (2022) examined 99 subscription business cases and found that consumers who previously perceived products as utilitarian under regular one-time orders tended to view them as hedonic when offered subscription orders. Furthermore, Li et al. (2023b) found that subscription shoppers emphasized hedonic motives rather than utilitarian ones for longer interval subscriptions. These findings are consistent with Miao et al.'s (2022) prediction that customers who use subscription services may prioritize hedonic benefits over utilitarian aspects in their purchase decisions. Accordingly, we propose our second hypothesis regarding differences in the moderating effects of customer–seller relationship phases across purchasing methods.

H2. The effectiveness of anthropomorphic chatbots varies depending on customers' purchasing options: a low behavioral realism (competent) chatbot is more effective among one-time order customers while a high behavioral realism (warm) chatbot is more effective among subscription customers.

We expect the chatbots' positive effects to diminish over time as the novelty effect fades. For example, some studies have identified diminishing novelty effects in voice AI chatbots (Sun et al., 2019) and wearable smart devices (Fronczek et al., 2023). Additional studies have examined customers' repeat purchase behaviors by incorporating consumers' forgetting functions in purchase recommendation models to improve recommendation performance (Kong et al., 2018; Zhang et al., 2024). Therefore, as the novelty of chatbots' recommendations wanes over time, their impact on customers' purchase intentions may decrease. Few studies have investigated chatbot effectiveness over different timeframes; for example, Roy and Naidoo (2021) examined the effectiveness of chatbots' human-like conversational style on users' time orientation, finding that present-oriented users responded better to warmth, whereas future-oriented users preferred competence. Accordingly, we propose the following hypothesis regarding the effects of anthropomorphic chatbots on customer behavior across different

timeframes.

H3. The positive effect of anthropomorphic chatbots on product purchases declines over time.

Fig. 1 outlines this study's theoretical framework. To our knowledge, this study is the first empirical examination of the effectiveness of anthropomorphic chatbots that combines form and behavioral realism through a real-world field experiment. Furthermore, this study examines empirical evidence on the moderation effect of chatbot anthropomorphism across customer–seller relationship phases—namely, one-time and subscription orders over varying timeframes. The results provide significant implications for business practices.

3. Research method

3.1. Experiment design

This study conducted a field experiment with a Japanese e-commerce direct-to-customer brand, "akyrise" (https://es.akyrise.jp/), which sells more than 40 cosmetic products, including facial soap and moisturizing cream. In recent years, cosmetics and beauty products have become accessible and popular on e-commerce sites. Customers can purchase these cosmetics through one-time or subscription orders. The field experiment was conducted over 62 days between June and August 2022. During this period, an annual summer campaign offering unique cosmetic products at discounted prices was conducted to attract more visitors. A specially designed webpage was created for the summer campaign to implement experimental chatbots. Notably, the chatbots in the experiment were not AI-based; however, they had basic rule-based functions to respond to customers' requirements.

The experiment used four types of chatbots, characterized by a combination of two visual appearances (human-like or cartoon) and two response styles (warmth or competence); each visitor was randomly assigned to one of the four chatbots. Fig. 2 shows two examples: cartoon/competent (low form realism/low behavioral realism) chatbot and human-like/warm (high form realism/high behavioral realism) chatbot. The left panel depicts the competent chatbot, which delivers a brief message, "The summer campaign has started." In contrast, the warmth chatbot shown in the right panel begins with a seasonal greeting before introducing the summer campaign. Both chatbots then prompt users to select topics for further inquiry via the chatbot interface. Fig. 2 outlines the interaction flow for a user selecting the option to learn more about "items related to skincare concerns." Differences in response style are also reflected in the responses to this query—the warmth chatbot responds to user's concerns with sympathy and empathy. Miao et al. (2022) conceptualized social content as a key element that enhances chatbots' interactivity in terms of behavioral realism.

In contrast to purely task-oriented chatbots (e.g., providing informational responses), friendly chatbots convey emotional and empathetic responses as social content. Some studies have employed the stereotype content model as their behavioral realism elements (Kull et al., 2021; Li et al., 2023a; Roy and Naidoo, 2021); thus, this study also adopts the warmth–competence dimension, which is essential in customer experience and loyalty management (Bennett and Hill, 2012; Scott et al., 2013). A preliminary test involving 88 participants was conducted to validate the differential impressions created by these response styles. The pre-test results revealed that the warmth chatbot was perceived as significantly warmer than the competent chatbot (M[warmth chatbot] = 3.21, n = 43; M[competent chatbot] = 2.87, n = 45; p = .027; Cohen's d = 0.65). In contrast, no significant difference was found in perceived message length (M[warmth chatbot] = 3.57; M[competent chatbot] = 3.48; p = .289; Cohen's d = 0.14). These results confirmed that users could perceive the distinct response styles as intended.

3.2. Data

During the experiment, the chatbots were displayed in 33,996 sessions on the e-commerce site to investigate the relationship between chatbot interaction styles and customers' subsequent purchase behavior. The chatbot usage history and registered customer data, including purchase history and demographics, were stored in separate data sources; therefore, we linked them using URLs, IP addresses, timestamps, and shopping site membership IDs, enabling us to identify registered customers who used the experiment chatbot in 5,922 sessions. The analyses were restricted to registered customers with the retailer's membership IDs so that their purchase histories could be used in the analysis; this reduced the total sample size.

Table 2 compares the number of impressions and usages for each chatbot pattern in the original and analysis datasets. The "usage" columns refer to sessions where at least one interaction occurred between the chatbot and the visitor. Neither dataset reveals significant differences in usage rates across all chatbot patterns.

Table 3 describes and summarizes the dependent and control variables, including additional demographic variables, for the total sample and for each chatbot condition. In particular, the membership rank variable represents the e-commerce site's membership levels, determined by each customer's cumulative purchase amount and membership duration. As customers progress through the ranks, they gain access to additional benefits such as free shipping and higher reward points for purchases. Comparing the control variables across chatbot patterns indicates that the RCT was successful and that there is no systematic bias among the experimental groups.

3.3. Model

A regression model is constructed using the quantity of products customers purchased via the chatbot as the dependent variable in the short and long term. This approach allows us to analyze how the 2 × 2 anthropomorphic chatbot patterns affect customer purchase behaviors by estimating coefficients for all four conditions relative to non-chatbot users, the baseline reference group. We observe the short-term quantity of purchases made on the day the customer used the chatbot as well as the long-term quantity of purchases made by the same customer within 1, 2, 3, and 4 weeks after using the chatbot. The cosmetic e-commerce retailer also offers two purchasing options: one-time and subscription orders; therefore, a multivariate Poisson log-normal regression analysis (Aitchison and Ho, 1989) is employed to allow for a non-diagonal, grouped variance–covariance structure across five time lags and two order options.

The number of purchases (y_{stk}) in session s with time lag t for order type k is modeled as follows.

$$\begin{aligned} &y_{stk} \sim Poisson(\lambda_{stk}), \log(\lambda_{stk}) = \boldsymbol{X}_s^{\top}\boldsymbol{\beta}_{tk} + \varepsilon_{stk}, \\ &\boldsymbol{\varepsilon}_s = \{\varepsilon_{s11}, \varepsilon_{s12}, \varepsilon_{s21}, \ldots, \varepsilon_{s52}\} \sim MVN(\mathbf{0}, \boldsymbol{\Sigma}) \end{aligned} \tag{1}$$

The dependent variable represents the quantity of products purchased by customers who visited the retailer's e-commerce site where the experimental chatbot was installed.[1] The time lag index, $t = 1$, represents the quantity of purchases until midnight on the following day when the chatbot was used[2]; $t = 2, 3, 4, 5$ represents the number of purchases within 1, 2, 3, and 4 weeks, respectively, from the day the customer used the chatbot. The index for the order option (k) takes the values $k = 1, 2$, representing the number of one-time order purchases and subscription order purchases, respectively.

[1] The main findings of this study remain robust even when using purchase amount instead of purchase quantity. The model results with purchase amount as the dependent variable did not differ significantly from those presented in the following section.

[2] The number of purchases made by midnight on the day the customer used the chatbot may not accurately reflect purchases by customers who visited the site late at night; therefore, we continued to observe them until the following day.

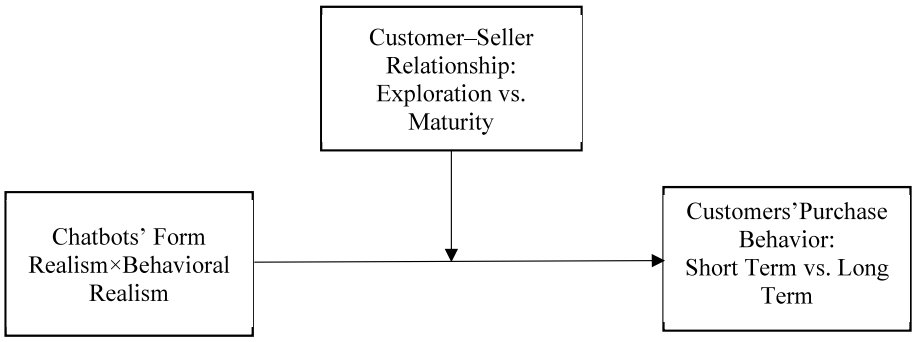


Fig. 1. Theoretical framework.

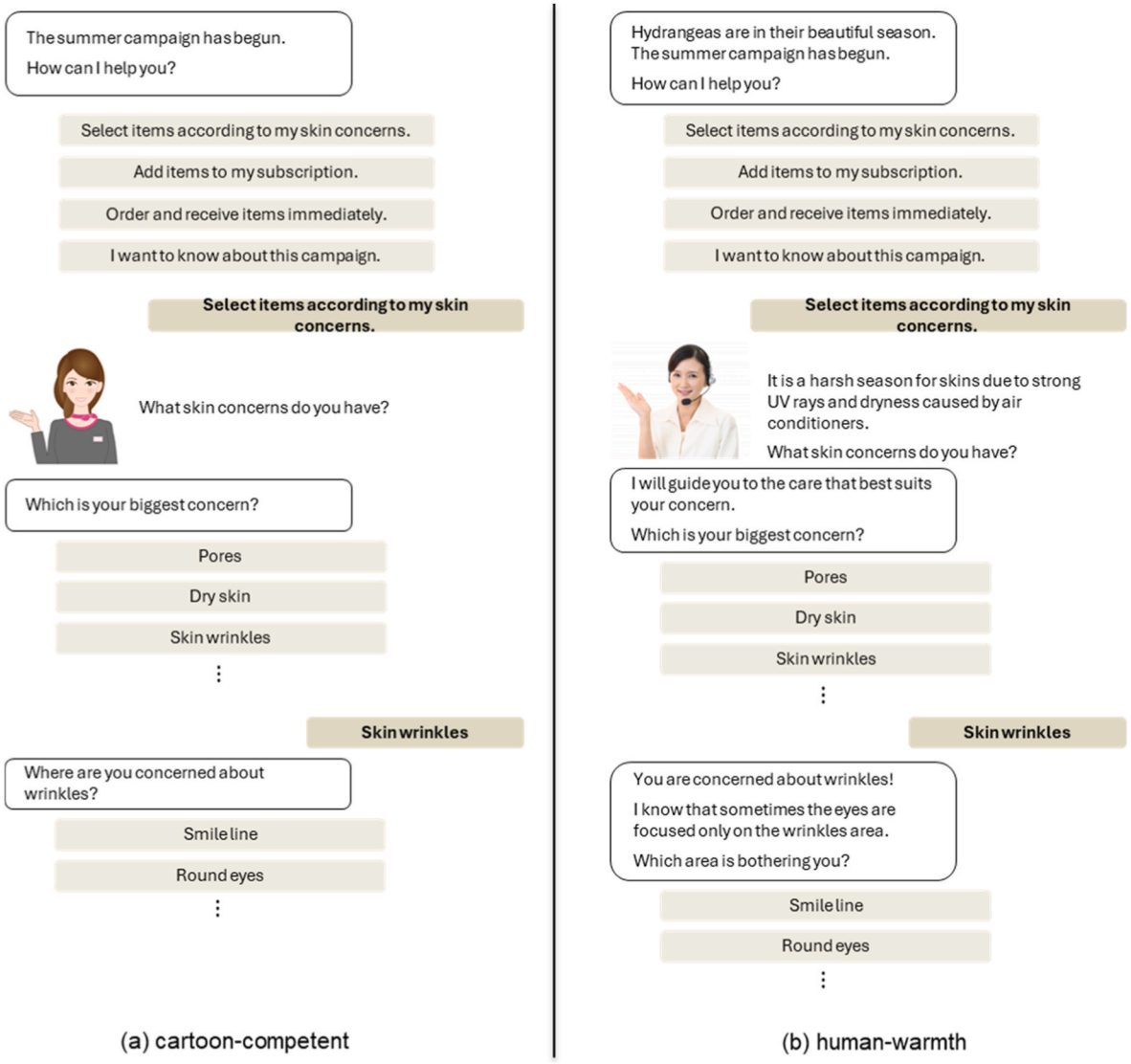


Fig. 2. Samples of the chatbot patterns.
Note: The left panel shows an interaction with a cartoon/competent (low form/low behavioral realism) chatbot and the right panel shows that with a human-like/warmth (high form/high behavioral realism) chatbot. The conversations in the figure are translated from the original Japanese.

Table 2
Number of impressions and uses for chatbot patterns.

Chatbot pattern	Original dataset			Analysis dataset		
	Impressions	Usage	Usage rate (%)	Impressions	Usage	Usage rate (%)
1. Human-like/Warmth	8,505	871	10.2	1,473	141	9.6
2. Human-like/Competent	8,454	909	10.8	1,488	149	10.0
3. Cartoon/Warmth	8,530	921	10.8	1,446	140	9.7
4. Cartoon/Competent	8,507	954	11.2	1,515	163	10.8
total	33,996	3,655	10.8	5,922	593	10.0

Note: Usage rates are calculated as usage/impressions.

This regression model assumes that the coefficients differ by time lag and order type, reflecting H2 and H3. The error terms, ϵ_{stk}, are neither independent nor identically distributed. Instead, they are grouped into a vector, $\boldsymbol{\epsilon}_s$, that integrates all time lags and order types, following a multivariate normal distribution. The covariance matrix, $\boldsymbol{\Sigma}$, represents correlations among order types and time lags. Model estimation was conducted using the *PLNmodels* package in R.

4. Results

This section presents the estimation results for four anthropomorphic chatbot patterns, separately for subscription and one-time order purchases. Log likelihood and Bayesian information criterion (BIC) of the estimated model are −42,377.7 and −43,311.5, respectively. In addition, an extended model examines the moderation effect of customer–-seller relationships to test H2.

4.1. Subscription order purchases

Table 4 presents the overall results of the Poisson regression model estimated for subscription order purchases. Fig. 3 illustrates the changes in the estimated effects over time lags for each chatbot pattern from Table 4. It also shows the comparison test results between each pair of chatbot patterns with the Wald test for joint hypotheses (e.g., testing the null hypothesis: $\beta_{1tk} = \beta_{2tk}$); Table A1 of the Appendix presents the detailed comparison test results.

The short-term effect of subscription purchases on the day of chatbot use shows a significantly positive effect only for the cartoon/warmth chatbot. This outcome supports H1 derived from the expectancy-disconfirmation theory. In contrast, the long-term effects of the cartoon/warmth chatbot do not differ significantly from those of other patterns after a one-week lag. This result suggests that, overall, the warmth chatbots consistently demonstrated significant effects. This finding aligns with H2, which hypothesized that high behavioral realism is superior during the maturity phase of subscription purchases. Hedonic responses, as reflected by warmth interactions, are likely to stimulate purchase intentions among subscription customers. The other types of chatbots initially showed no significant effect; however, over time, their effects became positive for subscription purchases, and there were no notable declines thereafter. No clear evidence of long-term decay was

Table 3
Summary of statistics.

	Total (n = 5,922)		1. Human-like/Warmth (n = 1,473)		2. Human-like/Competent (n = 1,488)		3. Cartoon/Warmth (n = 1,446)		4. Cartoon/Competent (n = 1,515)	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Dependent Variables										
One-time order										
within a day	0.37	0.89	0.37	0.88	0.38	0.94	0.38	0.88	0.34	0.86
within a week	0.46	0.97	0.45	0.94	0.47	1.01	0.47	0.96	0.44	0.97
within 2 weeks	0.51	1.02	0.51	0.98	0.52	1.05	0.51	1.00	0.51	1.03
within 3 weeks	0.56	1.07	0.55	1.02	0.56	1.10	0.55	1.06	0.58	1.10
within 4 weeks	0.60	1.12	0.59	1.08	0.59	1.12	0.59	1.12	0.63	1.17
Subscription order										
within a day	0.05	0.40	0.07	0.45	0.04	0.33	0.05	0.36	0.05	0.43
within a week	0.32	1.03	0.36	1.09	0.28	0.94	0.29	1.00	0.34	1.10
within 2 weeks	0.54	1.31	0.59	1.35	0.50	1.24	0.51	1.27	0.57	1.39
within 3 weeks	0.72	1.46	0.76	1.49	0.71	1.42	0.68	1.43	0.74	1.50
within 4 weeks	0.90	1.58	0.94	1.60	0.90	1.55	0.86	1.58	0.90	1.59
Control Variables										
Demographic variables										
MembershipRank(0/1 index)	0.62	0.49	0.63	0.48	0.62	0.49	0.60	0.49	0.63	0.48
Purchase365(Quantity for a year)	5.02	3.43	5.02	3.60	5.03	3.38	4.90	3.28	5.12	3.46
Subscription(0/1 index)	0.65	0.48	0.64	0.48	0.65	0.48	0.67	0.47	0.65	0.48
MailMarketing(0/1 index)	0.80	0.40	0.80	0.40	0.80	0.40	0.80	0.40	0.81	0.39
DirectMail(0/1 index)	0.85	0.36	0.85	0.35	0.84	0.37	0.85	0.36	0.87	0.34
Age*	51.27	8.31	51.45	8.51	51.04	8.20	51.09	8.00	51.48	8.52
Months since first purchase*	51.54	60.74	51.76	60.88	50.13	59.45	51.75	61.53	52.52	61.15
Session-related variables										
Mobile(0/1 index)	0.90	0.30	0.90	0.30	0.91	0.29	0.90	0.29	0.89	0.31
Holiday(0/1 index)	0.30	0.46	0.31	0.46	0.30	0.46	0.32	0.47	0.29	0.45
Evening(0/1 index)	0.43	0.50	0.40	0.49	0.44	0.50	0.42	0.49	0.44	0.50
Response(Quantity for a session)	0.30	1.21	0.26	1.01	0.28	1.12	0.31	1.24	0.36	1.41
Promotion(0/1 index)	0.22	0.41	0.23	0.42	0.23	0.42	0.20	0.40	0.20	0.40
Trend(Days)	27.90	19.80	28.42	19.80	28.64	19.73	27.79	19.41	26.73	20.01

Note: “Age” and “Months since first purchase” are not used as explanatory variables in the analyses. The values of “Age” are calculated excluding 220 missing values (3.7 % of the total sample).

Table 4
Estimation of the Poisson regression model for the number of subscription orders.

Covariate	Within a day			Within a week			Within 2 weeks			Within 3 weeks			Within 4 weeks		
	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.
Intercept	−9.478	0.402	***	−6.139	0.150	***	−5.184	0.118	***	−4.451	0.101	***	−3.746	0.091	***
Human_Warmth	−0.614	0.412		0.603	0.135	***	0.710	0.103	***	0.778	0.091	***	0.499	0.081	***
Human_Competent	−0.302	0.479		0.818	0.144	***	0.561	0.114	***	0.567	0.102	***	0.391	0.088	***
Cartoon_Warmth	0.815	0.395	*	0.917	0.150	***	0.719	0.115	***	0.882	0.098	***	0.695	0.085	***
Cartoon_Competent	0.036	0.414		0.248	0.177		0.595	0.122	***	0.866	0.100	***	0.602	0.088	***
MembershipRank	−0.712	0.163	***	−0.123	0.072	.	−0.047	0.053		−0.104	0.045	*	−0.109	0.040	**
Purchase360	0.236	0.018	***	0.216	0.007	***	0.194	0.006	***	0.201	0.005	***	0.187	0.004	***
Subscription	2.613	0.256	***	2.885	0.101	***	2.764	0.080	***	2.498	0.065	***	2.334	0.057	***
MailMarketing	0.338	0.196	.	−0.196	0.062	**	0.066	0.050		0.273	0.045	***	0.243	0.040	***
DirectMarketing	0.319	0.209		0.037	0.072		0.059	0.056		−0.033	0.048		0.027	0.043	
Mobile	−0.821	0.166	***	−0.646	0.065	***	−0.348	0.052	***	−0.216	0.047	***	−0.173	0.043	***
Holiday	−0.785	0.151	***	−0.311	0.055	***	−0.243	0.041	***	−0.171	0.036	***	−0.125	0.032	***
Evening	−0.106	0.126		−0.267	0.048	***	−0.252	0.036	***	−0.250	0.032	***	−0.220	0.028	***
Response	0.022	0.073		−0.047	0.031		−0.062	0.023	**	−0.118	0.021	***	−0.037	0.017	*
Promotion	0.043	0.140		−0.040	0.058		−0.060	0.044		−0.021	0.038		0.046	0.034	
Trend	0.003	0.003		−0.005	0.001	***	−0.003	0.001	**	−0.003	0.001	***	−0.004	0.001	***

Note: The "Signif." column represents statistical significance: *** <0.001, ** <0.01, *<0.05, and .< 0.1.

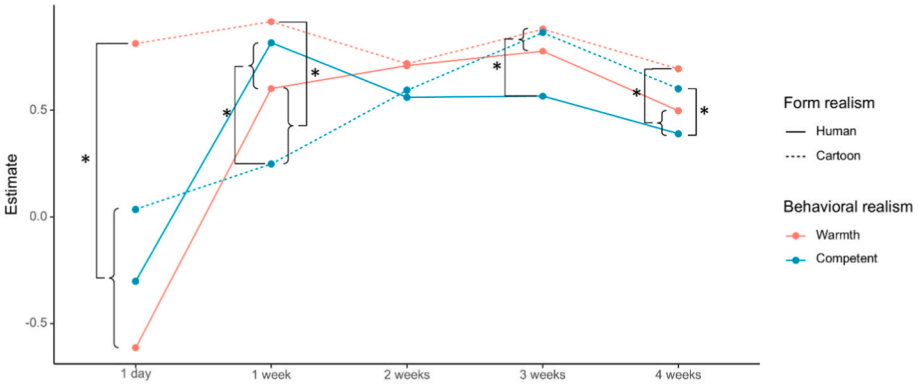


Fig. 3. Evolution of the effects of each chatbot pattern on the number of subscription orders.
Note: Asterisks (*) indicate statistical differences between these patterns at the 10 % level.

Table 5
Estimation of the Poisson regression model for the number of one-time orders.

Covariate	Within a day			Within a week			Within 2 weeks			Within 3 weeks			Within 4 weeks		
	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.
Intercept	−2.108	0.095	***	−1.990	0.088	***	−1.843	0.085	***	−1.734	0.082	***	−1.647	0.080	***
Human_Warmth	0.483	0.163	**	0.302	0.158	.	0.202	0.154		0.130	0.148		0.205	0.139	
Human_Competent	1.066	0.138	***	0.963	0.128	***	0.836	0.125	***	0.733	0.120	***	0.601	0.118	***
Cartoon_Warmth	0.966	0.150	***	0.891	0.137	***	0.735	0.135	***	0.662	0.128	***	0.503	0.127	***
Cartoon_Competent	1.069	0.152	***	0.980	0.140	***	0.886	0.135	***	0.791	0.128	***	0.690	0.124	***
MembershipRank	0.460	0.054	***	0.407	0.048	***	0.359	0.045	***	0.368	0.043	***	0.354	0.042	***
Purchase360	0.035	0.007	***	0.042	0.006	***	0.042	0.006	***	0.037	0.006	***	0.039	0.005	***
Subscription	−1.652	0.048	***	−1.605	0.043	***	−1.561	0.040	***	−1.495	0.038	***	−1.457	0.037	***
MailMarketing	−0.183	0.056	**	−0.052	0.052		0.012	0.049		0.025	0.048		0.078	0.046	.
DirectMarketing	0.348	0.065	***	0.391	0.061	***	0.412	0.058	***	0.443	0.057	***	0.424	0.055	***
Mobile	−0.239	0.062	***	−0.143	0.057	*	−0.124	0.055	*	−0.112	0.053	*	−0.078	0.052	
Holiday	0.248	0.047	***	0.188	0.043	***	0.162	0.040	***	0.157	0.039	***	0.146	0.038	***
Evening	−0.304	0.045	***	−0.254	0.040	***	−0.209	0.038	***	−0.169	0.036	***	−0.155	0.035	***
Response	−0.199	0.028	***	−0.204	0.027	***	−0.196	0.026	***	−0.165	0.024	***	−0.141	0.023	***
Promotion	−0.144	0.057	*	−0.083	0.050	.	−0.121	0.048	*	−0.128	0.046	**	−0.081	0.043	.
Trend	0.005	0.001	***	0.005	0.001	***	0.005	0.001	***	0.004	0.001	***	0.003	0.001	***

Note: The "Signif." column represents statistical significance: *** <0.001, ** <0.01, *<0.05, and .< 0.1.

observed; thus, H3 is not supported in the context of subscription purchases.

4.2. One-time order purchases

Table 5 summarizes the overall results of the Poisson regression model estimated for one-time order purchases, and Fig. 4 visualizes the changes over time and the significant differences between pairs of chatbot patterns (see detailed comparison test results in Table A2 of the Appendix). Most patterns show significant positive effects over time, indicating that customers who used the chatbot had higher purchase intentions than those who did not use it for one-time order purchases.

Furthermore, comparisons of chatbot patterns reveal that competent (low behavioral realism) chatbots have a greater impact compared with warmth (high behavioral realism) chatbots. In accordance with Proposition 11 of Miao et al. (2022), customers in the exploration phase tend to prefer cognitive responses from competent chatbots, which supports H2.

These results indicate that chatbots with a competence-based response style consistently outperformed others, regardless of form realism (cartoon or human-like chatbots). The cartoon/warmth chatbot also demonstrated competitive effects as no significant differences were found between the competent chatbots. This outcome suggests that while competent chatbots tend to have a more significant effect, the expectancy-disconfirmation theory also explains why the combination of a cartoon-like appearance and a warm response style positively impacts purchase experiences for customers, even in the exploration phase. The warmth chatbot's high behavioral aspect effectively mitigates customers' negative expectations from its low form (cartoon) appearance. This positive discrepancy also offsets the negative impact of the warmth response, thereby demonstrating effects comparable to those of competent chatbots. This outcome partially supports H1.

Nonetheless, the magnitude of these effects diminishes over time for all patterns, which aligns with our hypothesis that the chatbots' initial novelty effect may gradually wane (H3). These diminishing effects are nearly parallel across all patterns, indicating that the chatbot's novelty diminishes similarly in one-time purchases, regardless of the pattern.

4.3. Moderation effects by purchase order types

This subsection focuses on the moderating effects of customer–seller relationships on the varying effects of anthropomorphic chatbots to investigate whether the main results support H2. H2 posits that a low behavioral realism (competent) chatbot is more effective among one-time order customers, while a high behavioral realism (warm) chatbot is more effective among subscription customers. As the results from the main model estimation alone do not capture this moderation effect in detail, a modified Poisson regression model that includes an indicator variable for purchase order type (subscription order = 1; one-time order = 0) is employed in addition to the original model. Furthermore, we introduce interaction terms between four chatbot patterns and the order type indicator as explanatory variables, with their coefficients representing the moderation effects.

The estimation results are summarized in Table 6, and a detailed comparison of coefficients is provided in Table A3 of the Appendix. Log likelihood and BIC of the estimated model are −45,006.5 and −45,569.2, respectively. Table 6 shows that the moderating effects for the warmth chatbots are significantly larger than those for the competent chatbots. This finding indicates that customers who engage in subscription purchases tend to prefer warmth chatbots more than those who make one-time purchases, providing additional validating evidence for H2. Notably, in the short term (purchases made within a day), the moderation effects are significant only for the cartoon/warmth chatbot, indicating that the expectancy-disconfirmation effect (H1) strongly affects subscription customers, especially in the short term.

5. Discussion

5.1. Implications for the hypotheses

This subsection discusses the implications of our findings in relation to prior research results.

H1: Existing studies have found mixed results for chatbots' effectiveness in terms of the 2 × 2 anthropomorphic dimensions: form realism and behavioral realism (Blut et al., 2024; Yanxia et al., 2024). The findings in this study provide substantive evidence that chatbot designs combining low form realism (cartoon appearance) and high behavioral realism (warmth) positively influence customers' purchase behaviors in a real-world field experiment. The main results indicate that the most significant positive effect of the cartoon/warmth chatbot on subscription purchases and the competitive effect with competent chatbots on one-time purchases are estimated in both the short and long term. This outcome aligns with our expectations from the expectancy-disconfirmation framework, as outlined in Proposition 2 by Miao et al. (2022). In addition, the unique finding of this study is that the cartoon/warmth chatbot effects may apply only to the temporal effect of subscription order purchases, and their advantage diminishes as other chatbot patterns catch up, contrary to both the short- and long-term effects on one-time order purchases. Thus, the expectancy-dis confirmation theory may be effective only in the short term among subscription customers.

H2: In subscription purchases, customers are assumed to be in the maturity phase, where warmth chatbots with affective responses are

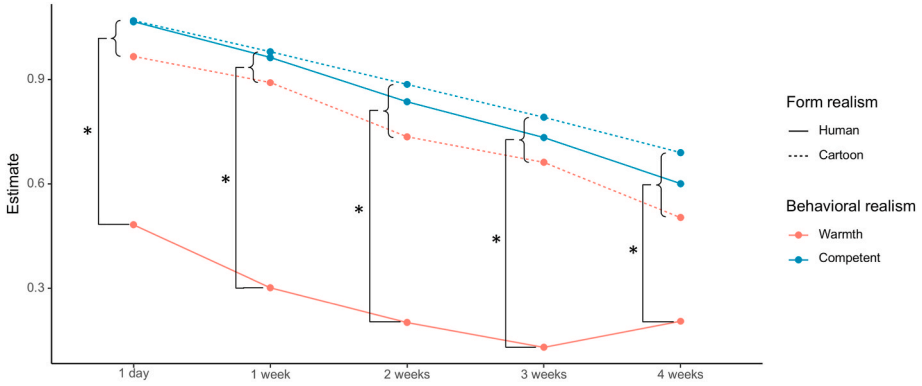


Fig. 4. Evolution of the effects of each chatbot pattern for the number of one-time orders.
Note: Asterisks (*) indicate statistical differences between these patterns at the 10 % level.

Table 6
Estimation of the Poisson regression model for the integrated dataset of subscription and one-time orders.

Covariate	Within a day			Within a week			Within 2 weeks			Within 3 weeks			Within 4 weeks		
	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.	Est.	S.E.	Signif.
Intercept	−2.578	0.090	***	−2.595	0.071	***	−2.536	0.063	***	−2.385	0.058	***	−2.248	0.055	***
Human_Warmth	0.097	0.162		−0.214	0.153		−0.339	0.149	*	−0.316	0.143	*	−0.301	0.133	*
Human_Competent	0.720	0.135	***	0.533	0.121	***	0.377	0.116	**	0.364	0.112	**	0.186	0.109	.
Cartoon_Warmth	0.541	0.147	***	0.353	0.128	**	0.173	0.125		0.219	0.118	.	0.023	0.117	
Cartoon_Competent	0.580	0.148	***	0.362	0.129	**	0.266	0.121	*	0.331	0.113	**	0.177	0.108	
MembershipRank	0.382	0.051	***	0.395	0.040	***	0.412	0.035	***	0.311	0.031	***	0.270	0.029	***
Purchase360	0.059	0.006	***	0.070	0.005	***	0.068	0.004	***	0.084	0.004	***	0.088	0.003	***
Subscription	−0.669	0.045	***	−0.327	0.036	***	−0.159	0.032	***	−0.099	0.029	***	−0.006	0.027	
MailMarketing	−0.105	0.053	*	−0.098	0.040	*	0.021	0.035		0.126	0.033	***	0.142	0.030	***
DirectMarketing	0.249	0.062	***	0.248	0.046	***	0.248	0.040	***	0.202	0.036	***	0.210	0.034	***
Mobile	−0.524	0.058	***	−0.455	0.043	***	−0.383	0.038	***	−0.342	0.035	***	−0.279	0.033	***
Holiday	0.055	0.045		0.021	0.034		0.019	0.029		0.038	0.026		0.042	0.024	.
Evening	−0.296	0.042	***	−0.253	0.031	***	−0.231	0.026	***	−0.222	0.024	***	−0.198	0.022	***
Response	−0.101	0.026	***	−0.083	0.019	***	−0.080	0.017	***	−0.094	0.015	***	−0.054	0.013	***
Promotion	−0.066	0.053		−0.023	0.038		−0.062	0.032	.	−0.046	0.029		0.018	0.027	
Trend	0.005	0.001	***	0.003	0.001	***	0.003	0.001	***	0.001	0.001	*	0.000	0.001	
OrderType	−2.242	0.069	***	−0.704	0.036	***	−0.186	0.031	***	0.091	0.028	**	0.296	0.026	***
HumanWarmth_OrderType	0.054	0.392		1.095	0.187	***	1.296	0.169	***	1.249	0.159	***	1.060	0.147	***
HumanCompetent_OrderType	−0.638	0.468		0.444	0.159	**	0.328	0.145	*	0.245	0.135	.	0.372	0.128	**
CartoonWarmth_OrderType	0.723	0.296	*	0.697	0.168	***	0.690	0.151	***	0.706	0.138	***	0.842	0.132	***
CartoonCompetent_OrderType	0.176	0.320		0.257	0.170		0.601	0.140	***	0.666	0.127	***	0.673	0.119	***

Note: The "Signif." column represents statistical significance: *** <0.001, ** <0.01, *<0.05, and .< 0.1.

preferred over competent chatbots. Conversely, in one-time purchases, customers are assumed to be in the exploration phase, where customers prefer competent chatbots that engage in cognitive responses over warmth chatbots. These findings confirm that the effects of anthropomorphic chatbots on customers' purchase behavior vary across order types representing customer–seller relationship phases, in accordance with Proposition 11 of Miao et al. (2022). This result contradicts the findings by Roy and Naidoo (2021), who showed that present-oriented (i.e., one-time order) users responded better to warmth, while future-oriented (i.e., subscription order) users preferred competence. This result can be attributed to the fact that customers who purchase subscriptions do not require detailed information and assistance from the chatbot. Conversely, one-time order customers may be in a trial-and-error stage, assessing the products' suitability and efficacy before transitioning to subscription purchases; thus, their experiences may be enhanced by evaluating product information from interactions with the low behavioral realism (competent) chatbot.

H3: For one-time purchases, the estimated effects of all anthropomorphism patterns diminish over time. This finding confirms the time-decaying novelty effect of the chatbot for one-time order purchases, consistent with the decreasing novelty observed for voice AI chatbots (Sun et al., 2019) and wearable smart devices (Fronczek et al., 2023). In contrast, no comparable decline is observed for subscription purchases across any chatbot pattern. Raj (1982) showed that high-loyalty consumers of a brand exhibited a longer carry-over effect of advertising on purchases compared with non-loyal consumers. Moreover, low-loyalty customers tend to be more price-sensitive to brand choice and respond more strongly to short-term promotional incentives (Krishnamurthi and Raj, 1991; Starr and Rubinson, 1978). Therefore, loyal customers are less likely to forget, leading to positive impacts that may persist over time across all chatbot conditions for subscribing customers.

5.2. Theoretical implications

This study makes several theoretical contributions. First, it integrates field-experimental evidence into the existing literature, which is dominated by laboratory-based studies (c.f. Table 1). Prior research has primarily relied on controlled experiments in hypothetical contexts; however, real-world field experiments with actual customers can offer robust insights into how chatbot anthropomorphism affects consumers' purchase behaviors, thereby strengthening the external validity of prior theoretical claims.

Second, this study confirms Proposition 2 of Miao et al. (2022). While a cartoon appearance of chatbots is less favorable than a human-like design, combining it with a warmth response style emphasizing high behavioral realism increases positive outcomes for customers' purchase behaviors. Contrary to the common assumption that more human-like designs always elicit positive user responses, this study's findings indicate that the expectancy-disconfirmation framework affects both one-time and subscription customers across timeframes. The results further show that the positive effect of cartoon/warmth chatbots is most substantial for short-term subscription purchases, while differences across the other chatbot conditions diminish over time. Oliver (1980) suggested that the disconfirmation effect may persist over time (Oliver, 1980, p. 466); however, our findings indicate that long-term effects may not fully hold in subscription contexts. Thus, future research should examine the enduring effects in longitudinal consumer–chatbot interactions.

Furthermore, this study provides longitudinal insights into customer behavior across relationship phases and timeframes. By analyzing repeated transactions over time, we demonstrate how the effects of chatbot anthropomorphism evolve from initial interactions to subsequent purchases. This dynamic perspective extends existing literature, which has typically focused on single interactions. Thus, this study highlights that consumers' perceptions and behavioral responses to chatbots are not static; novelty and forgetting effects may alter these

responses over time.

In summary, this study's contribution to the anthropomorphism literature is threefold. (1) It addresses a significant gap in prior research by examining the interaction between form realism and behavioral realism, an area where prior findings remain limited or inconsistent with Miao et al.'s (2022) framework. (2) Using a randomized controlled field experiment with actual customers, this study supplements the extant literature largely dominated by laboratory research, strengthening the external validity of anthropomorphism effects. (3) This research demonstrates that even chatbots with low form realism can enhance purchase behavior when paired with warm, human-like responses and further reveals a unique temporal pattern. These findings clarify the boundary conditions under which the expectancy-disconfirmation mechanism may operate across diverse purchasing types and different time lengths.

5.3. Managerial implications

This research also offers several practical implications for managers and practitioners. It provides practical guidance for e-commerce retailers on designing chatbot appearances and response styles and offers a framework for tailoring chatbot designs to best meet customers' varying needs. Specifically, the findings suggest that cartoon-like chatbots can be strategically effective when combined with warm, empathetic language. With rapid advancements in AI, particularly in large language models (LLMs), firms can easily develop AI chatbots with strong response capabilities by leveraging high-performance LLMs as back-end engines. Therefore, differentiating chatbots based solely on cognitive performance is becoming difficult. Against the backdrop of growing commoditization of high-performance AI chatbots, this study's findings highlight the importance of strategically designing form and behavioral realism. Regardless of how sophisticated a chatbot's response becomes, its visual appearance can influence consumers' expectations and purchase behavior. Similarly, behavioral realism (empathetic expressions and conversational tone) represents a customizable dimension independent of technological sophistication. In practice, OpenAI now offers multiple conversational tones in ChatGPT, enabling firms to tailor chatbots' personas (OpenAI, 2025).

Moreover, this study's results can help firms customize chatbot design across different stages of the customer–seller relationship. Retailers can strategically adjust chatbot appearance and communication tone—from competence-oriented interactions for new customers to more friendly designs for repeat buyers—to foster initial trust and long-term loyalty. While these effects diminish over time, especially among those making one-time purchase orders, firms should consistently update chatbots' appearance and interaction algorithms.

Beyond e-commerce chatbots, these insights can extend to the design of virtual influencers and AI avatars used in live commerce (e.g., Ozdemir et al., 2025; Zhou et al., 2024). The results may also apply to physical service robots, such as cat-shaped food delivery robots in restaurants (Nikkei, 2025) or humanoid receptionist robots (Cao, 2025), where appearance and behavioral realism similarly shape user expectations and interactions.

6. Conclusion and further research

The findings of this real-world field experiment provide empirical evidence clarifying the inter-relationship between chatbots' form and behavioral realism. Furthermore, this study elucidates the differences between one-time and subscription orders and the time-decaying effect on customers' purchase behaviors. While the study makes important contributions to the extant literature, it has several limitations. First, this study is limited to customers of the Japanese cosmetic e-commerce retailer, and Japanese culture may be more receptive to cartoon or character use in e-commerce sites compared with other regions. For example, Yanxia et al. (2024) reported that the positive effects of chatbot anthropomorphism on customers' attitudes and purchase behaviors were generally stronger in Eastern than in Western countries. This tendency may stem from the collectivist nature of Eastern cultures, in which consumers feel a stronger sense of social belonging toward anthropomorphized chatbots. Future research should examine whether similar effects hold across other countries and explore a broader range of product categories, including durable goods (e.g., automobiles) and services (e.g., car rentals). Second, restricting the participants to registered customers of the retailer may lead to sample selection bias. The dependent variable in this study was purchase behavior, and only registered members could make purchases on this e-commerce website; therefore, the analysis was limited by the available data. The field experiment in this study was implemented as an RCT, with participants randomly assigned to different chatbot conditions, thereby mitigating concerns about internal validity. However, the behaviors of non-customers were beyond the scope of this study, and generalizing the findings to broader consumers is a topic for future research.

Third, although the chatbots in this study were not AI-based but rule-based, at the time the experiment was designed, ChatGPT had only recently emerged. Due to the potential randomness and lack of consistency in AI-generated responses, maintaining strict experimental control was technically challenging. Nevertheless, the findings of this study on the chatbot's visual appearance and conversational style may apply to AI-driven chatbots. With the rapid advancement of AI technology, future studies could examine the effectiveness of state-of-the-art AI chatbots that provide fluent, voice-based responses. Finally, this research considered customers' membership traits as their characteristics; it remains critical to consider other variations in customer characteristics, such as digital literacy, chatbot usage, and personality traits (e.g., loyalty, risk perception, and trust). Additionally, future research should investigate other aspects of the theory of avatar marketing by Miao et al. (2022), including the moderating effects of different types of uncertainty. For example, researchers can explore high-involved categories (e.g., finance and healthcare services) vs. low-involved categories and the mediating influence of anthropomorphic design features through users' cognitive, emotional, and social perceptions.

CRediT authorship contribution statement

Mirai Igarashi: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hiroshi Onishi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation, Formal analysis, Conceptualization. **Osamu Sakai:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Data curation, Conceptualization. **Shohei Hasegawa:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1
Difference between the coefficients of chatbot patterns for the number of subscription orders

Covariate1	Covariate2	Within a day			Within a week			Within 2 weeks			Within 3 weeks			Within 4 weeks		
		Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.
Human_Warmth	Human_Competent	−0.313	0.589		−0.216	0.166		0.149	0.234		0.211	0.059	.	0.108	0.284	
Human_Warmth	Cartoon_Warmth	−1.429	0.002	**	−0.315	0.050	.	−0.009	0.943		−0.104	0.340		−0.196	0.043	*
Human_Warmth	Cartoon_Competent	−0.650	0.173		0.355	0.041	*	0.115	0.353		−0.088	0.402		−0.104	0.278	
Human_Competent	Cartoon_Warmth	−1.117	0.040	*	−0.099	0.539		−0.158	0.233		−0.315	0.007	**	−0.304	0.003	**
Human_Competent	Cartoon_Competent	−0.337	0.543		0.571	0.001	***	−0.034	0.796		−0.299	0.008	**	−0.212	0.034	*
Cartoon_Warmth	Cartoon_Competent	0.779	0.049	*	0.670	0.000	***	0.124	0.342		0.016	0.885		0.093	0.332	

Note: The “Signif.” column represents statistical significance: *** <0.001, ** <0.01, *<0.05, and .< 0.1.

Table A2
Difference between the coefficients of chatbot patterns for the number of one-time orders

Covariate1	Covariate2	Within a day			Within a week			Within 2 weeks			Within 3 weeks			Within 4 weeks		
		Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.
Human_Warmth	Human_Competent	−0.583	0.003	**	−0.661	0.000	***	−0.634	0.000	***	−0.603	0.001	***	−0.395	0.016	*
Human_Warmth	Cartoon_Warmth	−0.483	0.016	*	−0.589	0.002	**	−0.534	0.004	**	−0.531	0.003	**	−0.298	0.078	.
Human_Warmth	Cartoon_Competent	−0.586	0.003	**	−0.678	0.000	***	−0.684	0.000	***	−0.661	0.000	***	−0.484	0.003	**
Human_Competent	Cartoon_Warmth	0.100	0.566		0.072	0.654		0.100	0.524		0.071	0.638		0.097	0.516	
Human_Competent	Cartoon_Competent	−0.003	0.984		−0.017	0.915		−0.050	0.742		−0.058	0.688		−0.089	0.531	
Cartoon_Warmth	Cartoon_Competent	−0.103	0.555		−0.089	0.582		−0.151	0.337		−0.129	0.388		−0.187	0.206	

Note: The “Signif.” column represents statistical significance: *** <0.001, ** <0.01, *<0.05, and .< 0.1.

Table A3
Differences between the coefficients of chatbot patterns for the integrated dataset of one-time and subscription orders

Covariate1	Covariate2	Within a day			Within a week			Within 2 weeks			Within 3 weeks			Within 4 weeks		
		Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.	Diff.	P-val.	Signif.
Human_Warmth	Human_Competent	0.692	0.251		0.651	0.007	**	0.968	0.000	***	1.003	0.000	***	0.688	0.000	***
Human_Warmth	Cartoon_Warmth	−0.669	0.165		0.398	0.106		0.606	0.007	**	0.542	0.009	**	0.218	0.263	
Human_Warmth	Cartoon_Competent	−0.121	0.807		0.838	0.001	***	0.695	0.001	**	0.582	0.004	**	0.387	0.038	*
Human_Competent	Cartoon_Warmth	−1.361	0.013	*	−0.254	0.263		−0.362	0.078	.	−0.461	0.015	*	−0.470	0.009	**
Human_Competent	Cartoon_Competent	−0.814	0.145		0.187	0.413		−0.273	0.166		−0.421	0.021	*	−0.300	0.081	.
Cartoon_Warmth	Cartoon_Competent	0.547	0.198		0.440	0.060	.	0.089	0.662		0.040	0.828		0.170	0.332	

Note: The "Signif." column represents statistical significance: *** <0.001, ** <0.01, *<0.05, and .< 0.1.

Data availability

Data will be made available on request.

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